Project Charter

Problem Description

- Volleyball Video Analytics Software is an increasingly popular tool used by teams in the United States. It allows players and coaches to watch past games and practices to gain insights about potential tactical and technical enhancements. Users of this software can take advantage of the features that make the analysis process more efficient and accurate. For example, coaches can apply filters to analyze missed serves throughout the game.
- As the commercially-available solution used by most volleyball teams currently costs $10,000 per year, and is available only to the paying teams, not all the volleyball players have access to this tool. We intend to create a Volleyball Video Analytics model that would serve as a free resource for players and teams who may not have the resources to take advantage of the for-pay product.



Scope

1. What data science solutions are we trying to build?

- We propose a Semi-Supervised Computer Vision Model that will be able to annotate volleyball elements (*See: Problem Description*) from video clips provided by the user.
- The goals of this solution include: tracking the ball, accurately annotating volleyball elements, and dealing with occlusion happening during the play.
- The model will utilize supervised machine learning techniques, including Convolutional Neural Networks, Computer Vision Models (YOLOv8), and Computer Vision Libraries (OpenCV).

2. What will we do?

- Data Collection and Preparation:
  We will collect and annotate the videos of volleyball games. Then we will prepare the data to fit into the model (*See: Architecture*).
- Model Development:

We will train a Computer Vision Model on the dataset labeled by our team. We will fine-tune our model and add more training data for optimal results.

- Evaluation and Validation:
  We will evaluate the performance of our model using common Computer Vision metrics (*See: Metrics*).

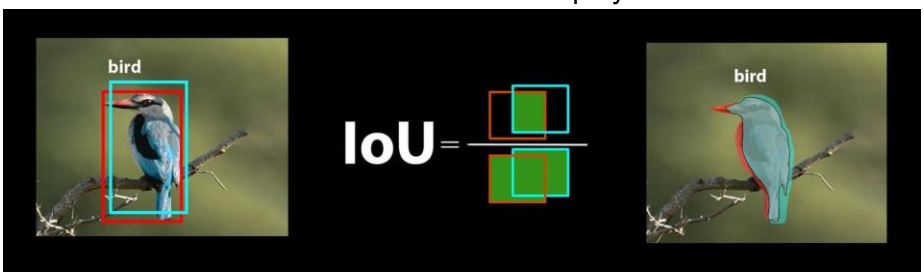
3. How is it going to be consumed by the customer?

- The consumer will be able to download the model from GitHub and let the model annotate their volleyball clip.

4. Limitations

- Our model will be trained primarily on the videos of women's college volleyball which might introduce bias
  - eg. towards players with ponytails or particular camera angle with the court.

Metrics

- IOU(Intersection over Union)(Area of Overlap/Area of Union) will be used to evaluate the correctness of the ball and the player's location detection.
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  - Definition
    - Comparison between the predicted bounding box and the ground truth bounding box. Calculated as the ratio of the area between the intersection of the two bounding boxes to the area of their union.
  - Implementation
    - Prediction
      - Run our model on testing videos to obtain the predicted bounding box.
    - Ground Truth
      - For each of the frames, there will be manual annotation of ground truth bounding boxes for the ball and the players with detected volleyball elements.
    - Calculation
      - For each frame of the video, calculate the IOU with the corresponding ground truth bounding box.
    - Threshold
      - IOU > 0.5 - True positive(TP) - The model correctly identifies the volleyball in each frame and the prediction box overlaps with most of the ground truth box.
      - IOU < 0.5 - False Positive(FP) - The model incorrectly identifies the volleyball in each frame, and the prediction box overlaps with less of the ground truth box.

                - False Negative(FN) - There is no explicit threshold, but it is the case when the model fails to detect the ball when it is present according to the ground truth data
                - True Negative(TN) - There is no explicit threshold and it is not applicable in object detection models, since the objective is to detect existing objects, rather than to detect non-existent objects.
            - Evaluation
                - Find the average IOU of all the frames. A higher average means better performance, while a lower IOU means there is more need to refine the model.
- Precision
    - $\frac{TP}{TP+FP}$
    - The proportion of correctly identified instances of a volleyball element out of all instances identified as a volleyball element by the model
        - A high precision indicates that the model has a low false positive rate and it doesn't mistakenly detect another object
- Recall
    - $\frac{TP}{TP+FN}$
    - The proportion of correctly identified instances of a volleyball element out of all instances of volleyball elements in the data.
        - A high recall indicates the model has low false negative and it is able to detect the volleyball elements when it is present in the data

Plan

- By 3/1 we intend to label 500 frames for Yolo and test the model.
    - If not, we will delegate one team member to increase the labeled dataset. (*See: Architecture; Yolo*)
- The First Deliverable is intended to be finished by 3/20.
    - Ball Tracking (*See: Architecture; OpenCV*)
- Our Second Deliverable is intended to be finished by 4/17.
    - Human Action Recognition
    - Labeled Volleyball Elements
- Final Deliverable and Demo to be presented on 4/26
    - Future Work
        - In the future, there's potential to continue our work by creating additional features such as detection of the quality of the volleyball element eg. good pass or bad pass.

Architecture

Data & Data Preparation

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- Primary Source:
    - Six recordings of College Volleyball games provided by a teammate's coach
        - All six videos are recorded from the one end of the court pointed towards the other end of the court

        - Three of which are located in the same court
        - Three others are located all at different courts
        - Each video is at least 1 hour in length

- Backup Source:
    - Hundreds of recordings of Professional Volleyball games provided by the youtube channel @MMGVolley
    - There are many different recordings with different courts, but there is at least 4 recordings, each at least 30 minutes, in the same court
- Preparation:
    - College Recordings
        -
    - Professional Recordings
    -
- The volleyball games videos will be manually annotated using minimum: 500 frames and prepared for Computer Vision Model Training.
- Video frames will be extracted using OpenCV (*See: OpenCV*) and split randomly into training and testing sets to ensure diversity of training data.

GPU

- Personal + Google Collab
    - During the early stages of our project, personal machines and the free version of Google Colab will suffice.
- Google GPU Cloud Host
    - Deep learning is a GPU-intensive endeavor, and with the use of the YOLO algorithm, it is even more necessary to seek a GPU cloud host for training on our data.
    - Using the NVIDIA T4 model - After researching our options, we agreed on the use of this GPU model as the good balance between high compute capabilities and reasonable price. (*See: Budget*)

YOLO

You Only Look Once is a common Machine Learning model used in Object Detection tasks.

- Labeling Optimization
    - We propose a semi-automated approach to labeling with the YOLO model that will allow us to gather more training data in a shorter amount of time.
- Efficient Object Detection
    - YOLO processes video frames through a single pass through a neural network, which enables real-time object detection which is ideal for video streams.
- High Accuracy
    - YOLO divides each of the video frames into grids, and predicts the bounding box and the class of the object in each of those grids, leading to accurate object localization and classification which is crucial in labeling the ball accurately in each frame of the volleyball video

OpenCV

Open Source Computer Vision Library is used for image processing, video frame extraction, and video analysis tasks.

- Using VideoCapture class to read a variety of video data types (i.e. AVI, mp4)
- Using the read() method, we can extract the frames of the selected video.

    - Extract frames from each of the videos at regular time based intervals
    - If the video has high frame rate, the regular interval would be longer
    - If the video has lower frame rate, the regular intervals would be shorter
- Error Handling:
    - Using functions in OpenCV, we can check whether each of the processes, from reading to extraction, is performed correctly.

Data Movement/Storage

- The video data is stored on google drive accessed by team members.
- We will potentially, move it to Google Cloud Platform after labeling, when the amount of labeled video frame gets too large
    - Since the video data is currently stored in Google Drive, Google Storage transfer service makes it easy to move our data.

Platform

- The client will be able to access the project through a GitHub repository.

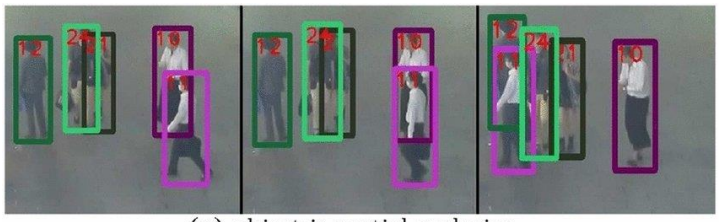
Communication

- Currently, the general roles are as follows:
    - Magda: Team lead, Feature Engineering, Model Building
    - Kat: Research, Labeling, Model Building
    - Jefferson: Feature Engineering, QA, Model Building
- There will be two meetings a week:
    - The first meeting will be early in the week and is a collaborative meeting to address what each team member will need to do ideally before the next meeting.
    - The end-of-the-week meeting will be a check-in to see how weekly progress went and what targets were hit and missed. We will then try to see if the missed targets can be solved in a group setting before the next meeting.
- To enhance communication and assure team member participation and satisfaction we have a text chat in which members can send feedback and updates on the project at any time to address quick fixes and prioritize during the end-of-week meeting.

Risks

- Budget
    - At this time we do not know the limit to our budget so we are asking for $250. This should cover our maximum projected costs based on the hourly rate.
    - If this is outside of budget we will look toward determining if we can save hours anywhere.
        - While alternatives exist, due to our intention of storing data on the Google Cloud and our familiarity with Google products we believe this will be the most cost-effective resource at this time.
- Labeling
    - Computer Vision Projects generally require large amounts of data for Model Training. We propose a solution to speed up the data annotation process. (*See: Yolo;Architecture.)*

- Occlusion
  - This problem is one of many Video Analytic models and currently has no foolproof solution. It does however have several proposed work around to optimize the accuracy surrounding these issues.
  - Below is a paper proposing using the tracked ball trajectory to estimate areas where the ball will appear once the period of occlusion ends. It claims when compared to similar models it surpassed them.
    - https://ieeexplore.ieee.org/document/7797015



(a) object is partial occlusion



(b) object is full occlusion